Ethics in AI

Dr. Rabab Ali Abumalloh

Ollscoil Teicneolaíochta an Atlantaigh

Atlantic Technological University

How AI uses our drinking water - BBC World Service
Copy link
BBC World Service is a British public broadcast service.
How thirsty is AI?
BBC NEWS | WORLD SERVICE
Watch on YouTube

AI Beyond the Screen

- Most people think of AI as “software” — but large AI models need physical infrastructure: data centers, cooling systems, energy.
- These have real environmental costs, including water usage.

Why Does AI Use Water?

a) Data Centers and Heat
- AI models run on powerful hardware that generates lots of heat.
- To keep servers operating safely, data centers rely on cooling systems.
- Many cooling systems use liquid cooling, which then must be cooled by water.

b) Potable Water Use
- Some data centers use clean drinking water (not just recycled or grey water) because impurities can cause corrosion, clogging, and bacterial growth in cooling equipment.

Scale of the Issue

- Over a billion AI chatbot interactions (like with ChatGPT) happen daily and each one contributes to this water usage.
- While individual water use per interaction is small, it adds up at scale.

Broader Environmental Impacts

a) Water Scarcity

- In places with limited clean water, data center demands compete with local communities, agriculture, and ecosystems.
- Reports mention concerns in several countries about water stress due to data center operations.

b) Energy & Carbon

- Water isn't just used for cooling — it's also a part of electricity production (especially fossil fuels), meaning AI's footprint connects to climate change too.

AI2027: Is this how AI might destroy humanity? - BBC World Service
Copy link
BBC World Service is a British public broadcast service.
Could AI eliminate humans in 10 years?
BBC NEWS | WORLD SERVICE
Watch on YouTube

What Is the “AI2027” Scenario?

- The video imagines a future where artificial intelligence becomes extremely powerful and risks harming humanity.
- It’s not a prediction with hard evidence, but a scenario raised by some researchers to spark discussion about long-term AI risks.

Why Do People Talk About AI “Destroying Humanity”?

There are three main reasons this kind of scenario gets attention:

a) Superintelligence Fears

Some theorists worry that if AI ever became smarter than humans in most tasks, its goals might not align with ours.

This idea is tied to long-term speculation about artificial general intelligence (AGI) — systems as capable as humans across a wide range of tasks.

Why Do People Talk About AI “Destroying Humanity”?

There are three main reasons this kind of scenario gets attention:

b) Media and Public Imagination

Stories about AI “destroying humanity” are dramatic and get attention — but they often mix fiction and fear with fact, so it’s important to separate evidence from hype.

Why Do People Talk About AI “Destroying Humanity”?

There are three main reasons this kind of scenario gets attention:

c) Preventive Discussion

Experts sometimes explore worst-case scenarios not because they believe them, but to encourage safety research and careful governance.

(These themes are commonly discussed in academic and media debates about future AI risk.)

What Are Realistic Concerns About AI?

While claims about AI wiping out humanity are speculative, there are serious, evidence-backed concerns:

a) Misuse and Harm (Today)

AI tools can be misused:

- Generating disinformation
- Biased decisions
- Automation that impacts jobs and privacy

These are grounded in current tech, not future speculation.

What Are Realistic Concerns About AI?

While claims about AI wiping out humanity are speculative, there are serious, evidence-backed concerns:
b) Security and Safety Research
Researchers study how advanced AI systems could behave unpredictably as they get more powerful. This is about ensuring alignment with human values, not necessarily “destruction.”
Safety experts want to make AI:

- Trustworthy
- Explainable
- Under human control

This is different from Hollywood apocalyptic scenarios.

How AI is changing the job market - What in the World podcast, BBC World Service
Podcast
Copy link
BBC World Service is a British public broadcast service.
Will AI take your job?
BBC NEWS | WORLD SERVICE
Watch on YouTube

How AI Is Being Used by Employers

The reporter explains that:

- Some companies use AI tools to filter applications and select candidates.
- In some organizations, AI is part of performance evaluation — employees may be expected to use AI tools as part of their work.

AI isn’t just replacing jobs — it's also reshaping how hiring and work are done.

Sectors and Workers Affected

AI tends to affect:

- Administrative and repetitive roles first.
- Entry-level positions that traditionally helped young people start career paths.
- Recruitment processes where machines filter candidates before a human ever sees an application.

Activity: Make a list of jobs that might be easily automated — and those that are harder to automate.

Sectors and Workers Affected

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

Case Examples from Around the World

The video includes examples from different regions:

- In some countries people report companies cutting junior roles.
- In India, some tech companies adopt AI but still need humans for oversight.

Sectors and Workers Affected

- Think of AI as a tool that can support work rather than replace it entirely.
- Focus on soft skills — critical thinking, communication, creativity — that are harder for AI to replicate.
- Combine technical knowledge with understanding of people and problems.

BBC WORLD SERVICE
AI-trained domestic robots: From science fiction to reality? - BBC World Service
Copy link
BBC World Service is a British public broadcast service.
Ready for the robots?
BBC NEWS | WORLD SERVICE
Watch on YouTube

What Are Domestic Robots?

A domestic robot is:

- A physical machine
- Operating inside homes
- Designed to assist with everyday tasks

Examples:

- Cleaning
- Organising objects
- Assistance and companionship

What Are Domestic Robots?

Why Are Domestic Robots Difficult?

Homes are:

- Unstructured
- Unpredictable
- Designed for humans, not machines

Challenges:

- Clutter
- Stairs
- Fragile objects
- Human and pet safety

What Are Domestic Robots?

Role of AI in Domestic Robots

AI allows robots to:

- Perceive their environment (vision, sensors)
- Learn from data
- Adapt behaviour over time

Without AI:

- Robots would only follow fixed rules
- No learning or flexibility

What Are Domestic Robots?

Ollscoil Teicneolaíochta an Atlantaigh
Atlantic Technological University

How Robots Learn Tasks

Robots are trained using:

- Human demonstrations
- Recorded movements and actions
- Large datasets

This is known as:

- Imitation learning
- Data-driven learning

Robot observes → learns → repeats task

What Are Domestic Robots?

Safety Concerns

Domestic robots must:

- Avoid harming people
- Detect humans and pets
- Respond safely to errors

Risk areas:

- Moving parts
- Gripping objects
- Operating near children

What Are Domestic Robots?

Privacy Concerns

Many domestic robots use:

- Cameras
- Microphones
- Sensors

Key questions:

- What data is collected?
- Where is it stored?
- Who can access it?

Homes = private spaces